

DOINGG@UNION.EDU

COURSE: CSC/BIO243: BIOINFORMATICS

SELF

Contents

Home 5

Announcements 5

Course Description 6

Schedule 7

Schedule 7

Resources 9

Python 9

Python Documentation 9

Gradescope: used to submit programming assignments 9

Google Colab Pro 9

Python Style Guide 10

Computer Labs 10

For help on your homework and projects, available to your are: 10

Syllabus 11

Grading 23

Assignments 25

HW 01 27

CSC/BIO 243: Assignment 1 29

Part 1 29

Part 2 29

HW 02 31

CSC/BIO 243: Assignment 2 33

HW 03 35

CSC/BIO 243: Assignment 3 37

Weekly Notes 39

Week 1 41

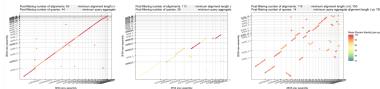
Notes from Week 1 43

Week 2 45

Week 3 47

About 49

Home



The course website for [CSC/BIO243](#), part of the Union College [CS curriculum](#). Here you can find our weekly schedule, assignments and resources.

Announcements

Welcome! You may want to bookmark this page for the term.

Office Hours

- Student/Office Hours:
 - Steinmetz 108B
 - Monday 2:00 pm - 3:00 pm
 - Thursday 2:00 pm - 4:00 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
 - by appointment for another time or over zoom

Course Description

Bioinformatics is the study of how information technology, computer science principles, and algorithmic techniques have affected and informed the study of biology in the 21st century (and vice versa). Specifically, the field of genomics (the study of the function of genes) has generated a tremendous amount of data to be analyzed. Bioinformatics brings to bear data management and analysis techniques found in the information processing field to discover pertinent knowledge in this sea of data that has applications in research and medicine.

In this course, you will learn about both biological and computer science concepts, how they interact with each other, and how they are used together to further research in genomics.

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4-3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1-5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1-6.1, 7.1-7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline

W	TOPIC	Reading	Due
8	Polymerase Chain Reaction	BDF Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	Project 2 Check-in
10	Gene Prediction	BFD Ch. 5, pp. 145-153	Project 2

Resources

Python

- Python 3.13.X: the software package needed to write and run Python programs
- <https://www.anaconda.com/download/success?reg=skipped>

Python Documentation

- Official Documentaion: <https://docs.python.org/3/index.html>
- Can *always* be used as a reference
- Can be searched for module-specific documentation e.g. turtle

Gradescope: used to submit programming assignments

- <https://www.gradescope.com/courses/1287277>
- Entry Code:G6J44N

Google Colab Pro

- Pro account sign-up: <https://colab.research.google.com/signup>
- Use your Union email
- A pro account is not needed for this course, just an option Union pays for

- Notes template: <https://colab.research.google.com/drive/15rPmKyQKCCbwsdDCpC39o8hczWZ-B5zV?usp=sharing>

Python Style Guide

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: <https://peps.python.org/pep-0008/>

Computer Labs

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

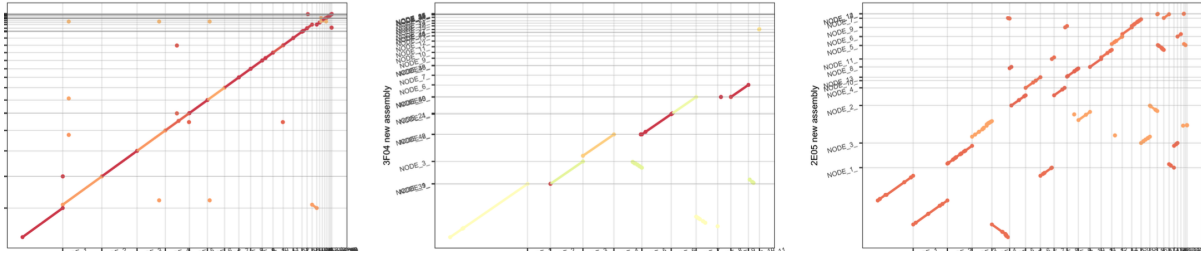
For help on your homework and projects, available to you are:

- CS Helpdesk (Olin 107, Sun–Thurs 7:00–9:00pm)
- Biology Backup (ISEC 316, Tues & Thurs 5:30–7:30pm)

Syllabus

Syllabus

CSC/BIO243 - Bioinformatics Syllabus



1 BIO/CSC 243: Bioinformatics

- Union College, Spring 2026
- Georgia Doing, PhD
- email: doingg@union.edu,
- office: Steinmetz 108B

2 COURSE BASICS

- Lecture: M/W/F 11:45 am - 12:50 pm
- Location: ISEC 070
- Student/Office Hours:
 - Monday 3:00 pm - 4:00 pm
 - Thursday 2:00 pm - 4:00 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app/gnYNtHjwwNxJ2UUN7>
 - by appointment for another time or over zoom

3 OVERVIEW

Bioinformatics is the study of how information technology, computer science principles, and algorithmic techniques have affected and informed the study of biology in the 21st century (and vice versa). Specifically, the field of genomics (the study of the function of genes) has generated a tremendous amount of data to be analyzed. Bioinformatics brings to bear data management and analysis techniques found in the information processing field to discover pertinent knowledge in this sea of data that has applications in research and medicine.

In this course, you will learn about both biological and computer science concepts, how they interact with each other, and how they are used together to further research in genomics.

3.1 Prerequisites and Course Format

The prerequisite for this course is BIO 205 (Current Topics in Molecular Biology) OR any introductory programming course CSC 10x. Because of this, it is expected that the student population of this course will consist of a mix of Biology and Computer Science students, each student bringing their own knowledge base, skill set, and perspective to the course. Given this diverse population, the first half of the course will concentrate on exposing students from one discipline to the fundamental concepts of the other (and vice versa) in order to gain a working knowledge of the field. Computer Science students will be taught the principles of Molecular Biology, and Biology students will be exposed to the essentials of Computer Science. In the second half of the course, we will assemble teams consisting of both types of students to work on a series of directed projects, the goal of which is to get students to creatively work in the inter- disciplinary fashion that is central to the field of Bioinformatics.

4 LEARNING OBJECTIVES

4.0.1 in Biology:

- basic Mendelian genetics
- the flow of genetic information from DNA to proteins
- molecular biological approaches and their applications
- genomic projects and comparative genomics

4.0.2 in Computer Science:

- algorithm design as a means of problem solving
- basic programming using the Python programming language

- data organization schemes, analysis techniques, and how they apply to biological applications
- further understanding of what computers can and cannot solve easily

4.0.3 in Bioinformatics:

- pairwise and multiple alignment matching of gene sequences
- polymerase chain reaction (PCR) techniques
- automated primer detection phylogenetic trees
- gene prediction

5 CLASS POLICIES AND GUIDELINES

The Computer Science and Biology Departments, and Union College as a whole, welcome all people, regardless of age, background, beliefs, ethnicity, gender identity, gender expression, national origin, religious affiliation, sexual orientation and any other differences, be they visible or non-visible. It's also important to recognize that institutional racism has prevented members of marginalized groups - especially black, indigenous and other people of color (BIPOC) - from fully participating in the fields of Computer Science and Biology.

You. Yes you, the one reading this syllabus. You belong here.

As an instructor, I will do my utmost to uphold these principles and to treat everyone with respect. As students in this class, I expect you to do the same since one person is not a community unto themselves. We're all in this together, so let's treat each other well. Lastly, I welcome feedback on issues we might discuss or ways that our class can be a more just one for all people.

6 COURSE MATERIALS

6.0.1 Texts (2-3 required)

- Claverie, J. and Notredame, C., *Bioinformatics for Dummies*, 2nd Edition, Wiley Publishing, 2007.
- Downey, A.B., *Think Python: How to Think Like a Computer Scientist*, Green Tea Press.
 - This book is free! Read it online at <https://alldowney.github.io/ThinkPython/>
- Kratz, R.F., *Molecular & Cell Biology for Dummies*, 2nd Edition, Wiley Publishing, 2020.
 - Only students who have not taken BIO 205 should buy this book.

We'll use the following websites and software throughout the course:

- The course website: a github-hosted site with assignments, materials and announcements
 - <https://georgiadoing.github.io/CSC243-S26/>
 - you may want to bookmark this to check it before and after each class
- Gradescope: used to submit programming assignments
 - <https://www.gradescope.com/courses/1287277>
 - Entry Code:G6J44N
- Google Colab: the cloud-based platform to write and run Jupyter Notebooks with python code
 - <https://colab.research.google.com/>

7 COURSE RESOURCES

The course website has a list and links to the readings and software resources that we are going to use. In class, we are going to use the Linux machines in ISEC 070. Outside of class you have a choice of hardware. You can access Google Colab via a browser on your own machine, install the software on your own computer (Python is freely available for Windows, Mac, and Linux) or you can work in one of the Computer Science labs. We have three spaces that you can use 24/7 using your ID card, except when classes are being held in them:

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

For help on your homework and projects, available to you are:

- CS Helpdesk (Olin 107, Sun-Thurs 7:00-9:00pm)
- Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm)

8 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: <https://www.union.edu/schaffer-library/ask-a-librarian>.

9 ACCOMODATIONS

It is the policy of Union College to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible, all discussions will remain confidential. You must provide reasonable notice and be in touch with Accomodative Services on the 2nd floor of Schaffer Library, if you have not already, if you wish to take advantage of extra time on exams or quizzes: <https://www.union.edu/accommodative-services>

10 GRADING

Some weeks I will give you *homework problem exercises*. You will complete these and submit them online, but they will not contribute to your final grade. Rather, they will prepare you for weekly *in-class quizzes*. There will be 2 *group projects*, and a *final exam*. Finally, class attendance, participation and engagement are critical components of the course.

- Participation & Engagment: 15%
- Quizzes: 30%
- Group Projects: 40%
- Final Exam: 15%

The final exam is not cumulative.

Programming projects must be turned in electronically via Gradescope. All programs will be done in the Python programming language. CS students will be expected to learn this language on their own during the first five weeks.

No late submissions will be accepted for any of the group projects. All exams and quizzes are closed-book, closed-notes.

Letter Grade Scale:

- A: 93-100; A-: 90-92
- B+: 87-89; B: 83-86; B-: 80-82
- C+: 77-79; C: 73-76; C-: 70-72
- D: 60-69; F: 0-59

11 ATTENDANCE

Class participation is a critical component of the course and attendance is mandatory. I realize that sometimes other things come up (interviews, athletic events, illness, emergencies etc.). In those cases, just let me know (in advance, if at all possible) that you are going to be absent. There may not always be the possibility of make-up credit for missed in-class material, but it is more likely for something to be arranged if you discuss your absence with me prior to class. No matter how much class you have missed, please do not come to class if you have an illness that is likely contagious, please email me to work out a reasonable solution.

If you do miss class, it is your responsibility to make up any material that you missed. Get notes from a classmate and make sure to complete all homework assignments due or assigned during the class that you missed. I will expect you to hand in all assignments by their due dates. You will not be able to make up missed exams, quizzes or in-class engagement points.

12 PARTICIPATION AND ENGAGEMENT

Following these guidelines will constitute positive participation and engagement and earn you up to a 0.5% of your final grade per class:

- Respect. This course is a space for rigorous and respectful debate. When confronting ideas and people different from you, lead with curiosity rather than judgement. We will critique ideas, not people. To protect our shared learning environment, you may not record, photograph, or share any part of our class sessions outside of our class.
- Engage. Show up to class on time and prepared. Show up ready to learn by completing the readings and exercises and checking the course website often. Engagement in class constitutes being present, respectful and engaged. Accessing off-topic websites or software, checking email or phones, *utilizing large language models* or otherwise attending to things not pertinent to the course is distracting to yourself and others.
- Embrace intellectual humility. Recognize that there are limitations to your knowledge and that some of your beliefs could be wrong. Be curious about your thinking and open to learning from others. This is hard, but so vital to your learning—we'll work on developing this throughout the term!
- Get help when you need it. If you are stuck or confused or lost, be proactive and get help. The best learners and thinkers can figure out when they are stuck and make a plan to get unstuck. Come to Office Hours (Steinmetz 108B), the CS Helpdesk (Sun-Thurs 7:00-9:00pm Olin 107) or Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm). Try reaching out to another student in the course who seems like they get it—they will probably be flattered you asked! Often the best person to explain something is the person who just figured it out.

13 ACADEMIC INTEGRITY AND “ARTIFICIAL INTELLIGENCE” USE

Union College recognizes the need to create an environment of mutual trust as part of its educational mission. Responsible participation in an academic community requires respect for and acknowledgement of the thoughts and work of others, whether expressed in the present or in some distant time and place. Matriculation at the College is taken to signify implicit agreement with the Academic Honor Code, available at: <http://muse.union.edu/honorcode>

It is each student’s responsibility to ensure that submitted work is their own and does not involve any form of academic misconduct. Students are expected to ask their course instructors for clarification regarding, but not limited to, collaboration, citations, and plagiarism. Ignorance is not an excuse for breaching academic integrity. Students are also required to affix the full Honor Code Affirmation, or the following shortened version, on each item of coursework submitted for grading:

Union College has an honor code as follows. That material will appear in every notebook you submit to me for grading.

As a student at Union College, I am part of a community that values intellectual effort, curiosity and discovery. I understand that in order to truly claim my educational and academic achievements, I am obligated to act with academic integrity. Therefore, I affirm that I will carry out my academic endeavors with full academic honesty, and I rely on my fellow students to do the same.

Scholastic dishonesty is misrepresenting someone else’s work as your own and will not be tolerated.

For this class in particular, we encourage working together during class time. If you missed something, talk to me, talk to your classmates or reach out via piazza or helpdesk. However ALL HOMEWORK must be completed individually. You are encouraged to:

- Talk about concepts in solutions
- Discuss ideas
- Look up online documentation, or examples

You MAY NOT:

- Share code
- Look at another student’s code
- Look up solutions to specific problems on the internet
- Copy or paste any text or code to or from a large language model

Here’s the bottom line: if you find yourself turning in work that looks substantially like the work of someone else, you should seriously examine whether you have crossed the line. If

you have any doubts, talk to me *before* turning in the assignment. Ultimately, I may choose to call you into office hours to explain the choices you made in solving a problem.

Your goal for discussions about any assignment should be that you come away with a better understanding of the problem and of possible ways to approach it so that you can then try out these approaches on your own. You should never leave a discussion with just an answer, without an understanding of how to arrive at that answer. A good general guideline for any discussion, or interaction with an AI model, is that you should not leave the discussion with anything written or typed.

14 CONTACTING ME OUTSIDE OF CLASS

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email (doingg@union.edu) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

15 ADDITIONAL RESOURCES

Mental Health and Campus Resources

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students (dos_office@union.edu) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are comfortable doing so and I will provide any resources I can. The Union College Persistence Fund can provide financial support in times of unexpected hardship to cover needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of support are outlined on the Dean of Students webpage: <https://www.union.edu/dean-students>.

16 TENTATIVE SCHEDULE

Check the course website often for any changes

W	TOPIC	Reading	Due
1	What is bioinformatics: what are computers and living cells good at?	Bioinformatics for Dummies (BFD) Ch. 1 & 2	Survey, Quiz 0
2	Algorithms & the Genetic Code	Think Python (TP) Ch. 1, 2, 3.4-3.7 & Molecular & Cell Biology (MCFD) Ch. 1, 6, 7	HW 1, Quiz 1
3	Modular Programming & Proteins	TP Ch. 5.1-5.7, 5.11, 6.1 & MCFD Ch. 16, 17	HW 2, Quiz 2
4	Loops & Sequencing	TP Ch. 6.1-6.1, 7.1-7.3, 7.7 & MCFD Ch. 19	HW 3, Quiz 3
5	Sequence Processing & Disease	BFD Ch. 7	HW 4, Quiz 4
6	Sequence Alignment	BFD Ch. 9, TP Ch. 8	HW 5, Quiz 5
7	Multiple Sequence Alignment	BFD Ch. 13	Project 1 Outline
8	Polymerase Chain Reaction	BFD Ch. 13, 9 cont.	Project 1
9	Phylogenetics	BFD Ch. 13 cont.	
10	Gene Prediction	BFD Ch. 5, pp. 145-153	Project 2

17 PROGRESS TRACKER

	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week
Learning Goals	1	2	3	4	5	6	7	8	9	10
Genetic Information										
Comparative Genomics										
Programming										
Algorithm Design										
Sequence Alignment										
Segeunce Interpretation										

18 ADDENDA

Any community agreed upon additions:

Grading

Table 2: Grading Scheme

Course Element	Grade Point Contribution	Notes
Participation	15%	see syllabus for details
Quizzes	30%	written
Group Projects	40%	coding
Final Exam	15%	written
total	100%	

Table 3: Letter Grade Scale

Letter Grade	Percent range
A	93 - 100
A-	90 - 92
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79
C	73 - 76
C-	70 - 72
D	60 - 69
F	0 - 59

Assignments

HW 01

CSC/BIO 243: Assignment 1

- Remember to check the schedule for readings: <https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html>
- Over the weekend
 - finish reading Bioinformatics for Dummies Ch. 1 & 2 and
 - begin the reading Think Python Ch. 1-3 or Molecular & Cell Biology for Dummies Ch. 1,6 and 7
 - you will want to be caught up on the reading since we will have our **first quiz next Wed of Fri**
 - I can answer any questions about materials from readings on Monday, ahead of the quiz

Part 1

- [Intro Survey](#)
- Read the Therac25 case study summary and one more detailed article (linked in [week1 notes](#), you can choose one of the two longer articles on the notes page or find another if you prefer. Read critically and carefully and come to class **Wednesday 4/1/26** ready to discuss.

Part 2

- Prepare a 5 minute introduction to a topic for which you bring background knowledge to this class
 - you can use slides, write on the board and/or have me post/print out

handouts

- Each topic will be presented at the beginning of the relevant class day:
[Topic Sign-up and Schedule](#)
- Based on the results on the survey, I have done my best to match students to topics they are already familiar with, HOWEVER, you should MOVE or ADD your name so you are comfortable with your topic
- Up to 2 students can sign up for each topic
- Questions that arise as you are preparing your topic are great reasons to come to office hours, I am happy to work with you!
- This presentation will earn you one quiz grade, “Quiz 0” (5% of your final grade)

HW 02

CSC/BIO 243: Assignment 2

- due Wednesday 4/15/26 by class time (11:45 am)

~~* due Monday 4/13/26 by class time (11:45 am)~~

- Remember to check the schedule for readings: <https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html>
- Finish the 'Practicing Functions and the Genetic Code' activity that was started in-class as Pair Programming
 - Bring your written answers to class next ~~Monday~~ Wednesday
 - You will submit your code, individually, to gradescope
 - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
 - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Development Environment (IDE) or a source-code editor with extensions (if this jargon doesn't mean anything to you, you are safe)
- Keep checking this page for any updates...

HW 03

CSC/BIO 243: Assignment 3

- due Wednesday 4/22/26 by class time (11:45 am)
- Remember to check the schedule for readings: <https://georgiadoing.github.io/CSC243-S26/02-Main/schedule.html>
- Finish the ‘[Practicing Conditionals, Loops and Protein Translation](#)’ activity that was started in-class as Pair Programming
 - Bring your written answers to class next Wednesday
 - You will submit your code, individually, to gradescope
 - Be sure to add any additional comments (header, description, pseudocode, inline) so the code is a final product rather than a draft
 - You do not need to work on a lab computer (although you always can!), but you should still be working in Idle or a plain text editor, not an Integrated Development Environment (IDE) or a source-code editor with extensions (if this jargon doesn’t mean anything to you, you are safe)
- Keep checking this page for any updates...

Weekly Notes

Week 1

Notes from Week 1

For Reading/HW

- [Therac25 Summary](#)
- [Therac25 Leveson](#)
- [Therac25 Silvis](#)

Monday in class

- [No More DNA](#)

Wednesday in class

- [Slides](#)
- [What are computers good at?](#)

Friday in class

- [Anatomy of a function](#)
- [Logging on to lab computers](#)
- [Pair Programming](#)
- [Rosalind Problems for Learning Python](#)
- [Rosalind Bioinformatics Practice Problems](#)
- [Rosalind Bioinformatics Textbook Problems](#)

Week 2

- Mon:
 - Therac25 discussion (see notes from [last week](#))
 - Pseudocode [practice activity](#)
- Wed: [Practicing Functions and the Genetic Code](#)
 - note, we will start the above activity in class on Friday
 - [demo code](#) of comments and functions
 - [demo code](#) to make random pairs of bio-cs names
- Fri: we will have a quiz on material up through Monday
 - we will talk more about what DNA is and how it is turned into protein and/or replicated
 - we will also begin the assignment above as a pair programming activity

Week 3

- Mon: [Practicing Functions and the Genetic Code](#)
 - note, the above activity was handed out in class last Friday
 - [demo code](#) of comments and functions
 - [demo code](#) to make random pairs of bio-cs names
- Wed: [Practicing Conditionals, Loops and Protein Translation](#)
- Fri:

About

References

